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Local contrastive loss with pseudo-label based self-training for semi-supervised medical image segmentation

Report Project

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1 Introduction

1.1 Motivation

Medical image segmentation is an important step in many healthcare applications. It helps doctors and researchers automatically identify and separate different structures, tissues, or organs in medical scans such as MRI or CT images. This process plays a key role in diagnosis, treatment planning, and in monitoring how a disease develops over time.

The traditional approach uses supervised deep learning methods. In simple terms, supervised means that the model learns from a large set of images where expert doctors have already labeled each part of the image. These methods learn to match each pixel of the image to a class label, which gives very good results when there are many accurate annotations available. However, labeling medical images is slow, expensive, and must be done by trained specialists. As a result, in many real cases there are not enough labeled images to properly train these models. Fully supervised methods therefore face important limitations such as overfitting and poor generalization, especially because the number of labeled images is small and the variability in anatomy, pathology, and acquisition settings is high.

To address these challenges, researchers have turned to semi-supervised learning, where both labeled and unlabeled data are used for training. The idea is that unlabeled data, which are much easier to collect, can still help the model better understand the structure of medical images. Over the years, several main strategies have been developed in this field :

- **Consistency regularization** : This method uses both labeled and unlabeled images. Labeled data are trained with the standard segmentation loss. For unlabeled images, the idea is to enforce stable predictions under small perturbations. We take the same image, apply two different augmentations (rotation, noise, brightness change), and pass both versions through the model. The model is then trained to make these predictions as close as possible. This helps the decision boundary avoid noisy regions and improves global stability, but it does not focus on fine pixel-level differences. As a result, important local anatomical details may still be lost.
- **Entropy minimization** : Here too, labeled images use the normal supervised loss. For unlabeled data, the goal is to make the model confident in its predictions by reducing the entropy of its output probabilities. Low entropy means that each pixel is clearly assigned to one class rather than staying uncertain. This encourages sharper predictions on unlabeled images. However, confidence does not guarantee correctness : if the model is unsure or biased, it can become very confident about wrong predictions, especially when unlabeled images are different from the labeled ones.
- **Self-training with pseudo-labels** : This is one of the most common semi-supervised strategies. The model is first trained on labeled data, then used to predict labels for the unlabeled images. These predictions, called pseudo-labels, are then used as training data for the next steps. Over several iterations, the model learns from its own predictions. This can work well when pseudo-labels are accurate, but if they contain errors, for instance, wrong organ boundaries or false class assignments, the model can reinforce these mistakes. This problem is called *confirmation bias*, and it often becomes one of the main limits of self-training.

All of these methods help use unlabeled data more effectively, but they still struggle to capture fine anatomical structures and remain robust when labels are noisy or uncertain. In parallel, another research direction called **self-supervised learning** has become popular for learning image representations without labels. Instead of using annotated data, the model is trained on *pretext tasks* that force it to understand visual structures by itself, such as predicting missing regions, reordering shuffled patches, or matching two augmented versions of the same image. Although these tasks are unrelated to segmentation, they teach the encoder to extract useful low-level and mid-level features (edges, textures, shapes), which can later be reused and fine-tuned for segmentation when labeled data is limited.

Among self-supervised methods, **contrastive learning** has shown particularly strong results. The idea is to pull together representations of samples that should be similar (e.g., two augmentations of the same image) and push apart representations of different samples. This creates a latent space where

similar anatomical patterns lie close together.

In medical imaging, contrastive learning is often used as a pretraining step before training a segmentation network. This improves generalization, especially when only a few annotated scans are available. However, most contrastive frameworks operate at a **global level**, comparing whole images or large patches. This is effective for classification, but insufficient for pixel-wise segmentation, where fine spatial details are essential : two adjacent pixels may look similar but belong to different organs, while two distant pixels may look different but represent the same tissue.

Because global contrastive losses do not model pixel-level relationships, the learned features often become coarse and inconsistent across small regions. Some recent works therefore try **pixel-wise contrastive learning**, where each pixel selects its own positive and negative samples. But without real labels, it is extremely difficult to decide which pixels are truly similar.

Many methods rely on simple heuristics such as spatial proximity, feature clustering, or pre-segmentation with super-pixels. However, these cues are not anatomically reliable. This issue is illustrated in Fig. 1. Super-pixels split the image into visually homogeneous regions, but these regions do not correspond to real organs : a single super-pixel may mix several tissues, while the same organ is often divided into many unrelated fragments.

Moreover, **each super-pixel receives an arbitrary ID (color)**. Two regions belonging to the same anatomical structure will have *different IDs*, while two regions with the same ID across patients may not correspond to the same organ at all. As a consequence, the model cannot recognize consistent anatomy : it treats unrelated fragments as positives and splits true organ regions into multiple unrelated negatives.

This leads to feature representations that lack clinical meaning, are inconsistent across patients, and remain highly sensitive to small variations in appearance.

To summarize, the main problems with current semi-supervised segmentation methods are :

- **Error accumulation in pseudo-labeling** : Once wrong pseudo-labels are introduced, they are reinforced during training, leading to confirmation bias and lower accuracy.
- **Simplified consistency objectives** : Global consistency regularization helps keep predictions stable but fails to preserve fine local details that are essential for precise segmentation.
- **Global contrastive learning** : Most contrastive objectives ignore pixel-level relationships and anatomical context, resulting in oversmoothed or inconsistent representations.
- **Weak supervision for unlabeled data** : Unsupervised similarity definitions rarely match true anatomical structures, limiting reliability.

These limitations show the need for a new method that can use unlabeled data effectively, preserve local anatomical precision, and avoid the error propagation caused by noisy pseudo-labels.

This paper proposes a framework that combines local contrastive learning with pseudo-label self-training to address these issues. The key idea is to design a local contrastive loss that works directly at the pixel level, instead of relying on global similarity. It makes pixels from the same class close in the feature space and pushes apart those from different classes. This creates a more structured and anatomically meaningful feature representation.

The model is trained in two stages. First, a basic segmentation network is trained using only the labeled data. Then, in a joint training phase, segmentation loss is applied on labeled samples, while the local contrastive loss is applied on both labeled and unlabeled images using pseudo-labels.

To reduce the impact of wrong pseudo-labels, the authors also add a confidence-based selection step that keeps only the most reliable regions for the contrastive loss.

These proposed method in the paper produces sharper boundaries and overall better segmentation performance compared to previous semi-supervised and contrastive methods that we mentioned.

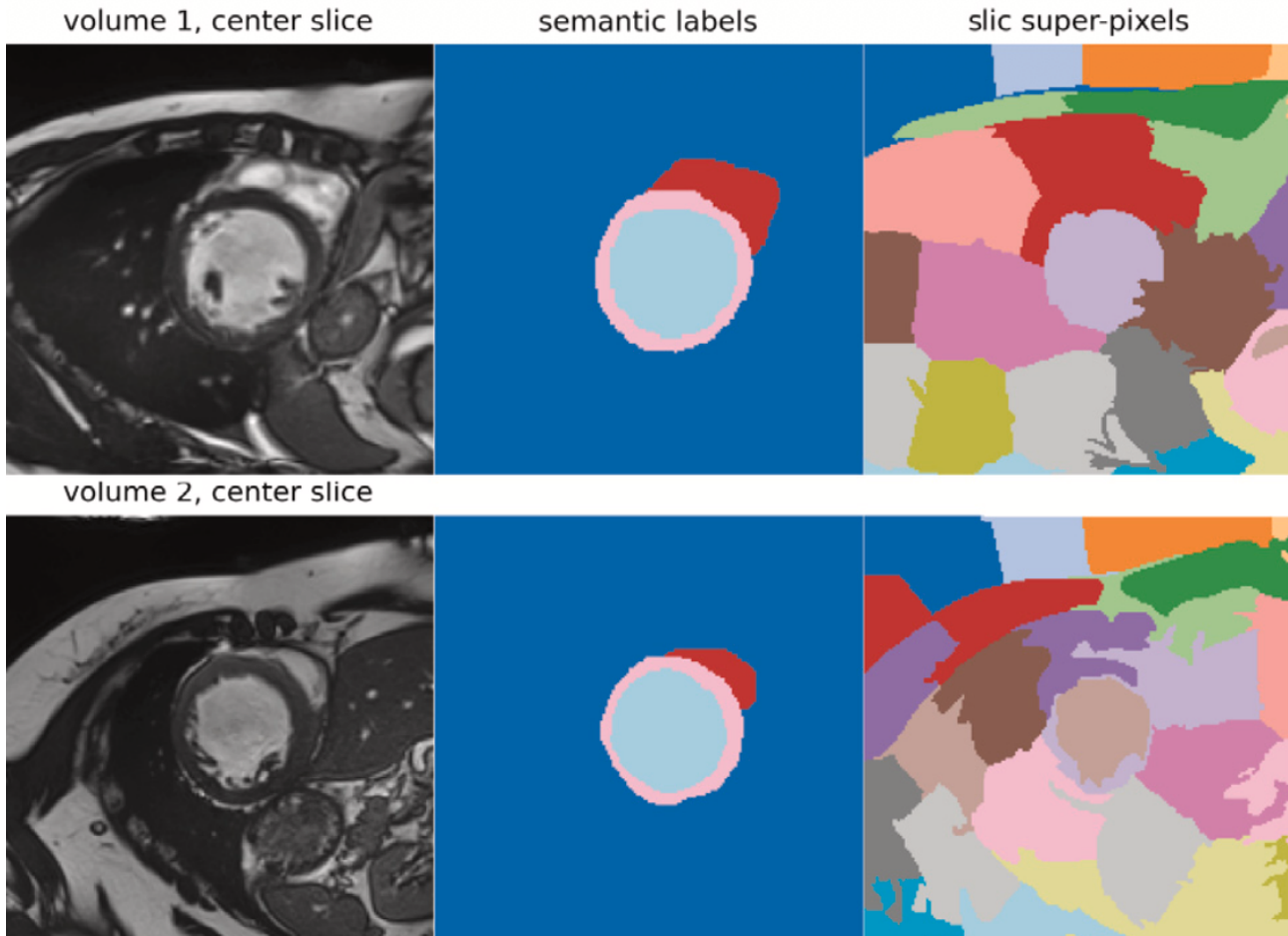


FIGURE 1 – Super-pixels do not align with real anatomy. Each color represents a super-pixel region with an arbitrary ID. Regions may mix tissues or split one organ into inconsistent parts, making them unreliable for pixel-wise contrastive learning.

1.2 Contributions

In this project, we make the following contributions :

- We reimplemented the whole method in PyTorch, including the U-Net backbone, the local contrastive loss and the two-stage training pipeline, while the original public code was written in TensorFlow.
- We ran again the experiments for the ACDC dataset from the paper and checked that our PyTorch implementation can reproduce their results on the public datasets they used.
- We extended the study to two new datasets : CHAOS, with five classes including background, and MSD Heart, with one organ class and background.
- We carried out a detailed analysis of the results : per-patient Dice scores on the validation sets, confusion matrices, t-SNE visualisations of the pixel embeddings, and visual inspection of representative slices.
- From these experiments, we observed that the local contrastive loss is especially helpful when there are several organ classes and the background is not too dominant (as in ACDC and CHAOS), where joint training improves over the supervised baseline. In contrast, when there is only one organ class with a very large background (as in MSD Heart), the contrastive loss often brings little or no gain, and the pure supervised baseline can even perform better.

During this project, Nicolas focused particularly on the methods section, used models, loss functions, training, and evaluation. Maxime completely rewrote the project in PyTorch, a library we are more familiar with than TensorFlow. He also worked on the introduction. Titouan worked on the experimen-

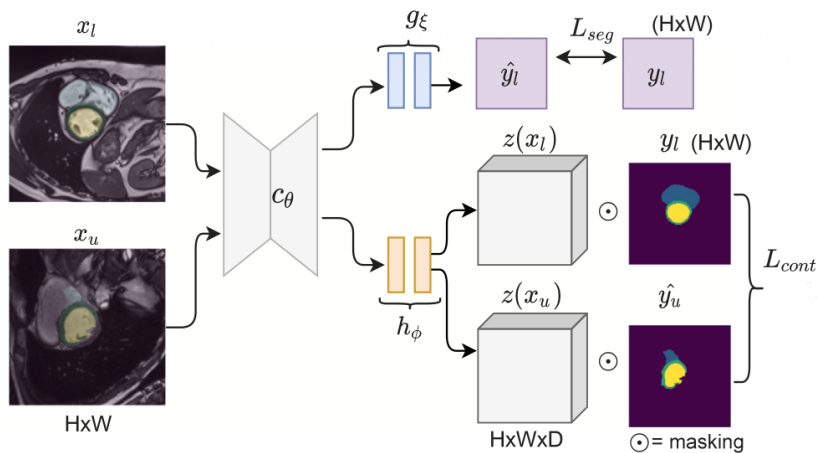


FIGURE 2 – Model architecture. Taken from [1]

tation part, adding a new dataset and exploring various visualizations, he also compared this methods with actual the state of the art one. Maxime and Titouan analyzed the results obtained by retraining the models on the different datasets.

2 Methods

2.1 Model and training

The authors of [1] propose a joint training semi-supervised approach with the introduction of a contrastive loss that is applied pixel-wise on pseudo-labels of unlabeled set and limited labeled set with ground truth labels.

The contrastive loss enforces pixels of a same class to have a similar representation but at the same time enforces them to have a different representation from pixel representations of other classes.

The model, presented in figure 2 consists in two separate network branches to compute segmentation and contrastive loss, where the segmentation branch only considers labeled set and the contrastive branch considers both labeled set and unlabeled set with their pseudo-labels. Pseudo-labels of the unlabeled set are generated during training with the current state of the segmentation network.

More precisely, we define the labeled set with X_L and its image-label pairs with (x_l, y_l) . The unlabeled set is defined with X_U , which is the set of pairs $(x_u, \hat{y}_u)$ where x_u is an unlabeled image and $\hat{y}_u$ is the associated pseudo-label. The complete dataset consisting in both labeled and unlabeled images is then $X = X_U \cup X_L$.

As said before, the model proposed in the paper is an encoder-decoder network c_θ followed by two separate branches : a segmentation network g_ξ and a network h_ϕ specific for training with contrastive loss. The algorithm proposed by the authors has two optimization steps :

1. The networks c_θ and g_ξ are first trained on the labeled set X_L by minimizing a segmentation loss L_{seg} :

$$\hat{\theta}^{(0)}, \hat{\xi}^{(0)} = \arg \min_{\theta, \xi} \frac{1}{|X_L|} \sum_{(x_l, y_l) \in X_L} L_{seg}(y_l, g_\xi(c_\theta(x_l)))$$

2. Joint training of the networks c_θ , g_ξ and h_ϕ is performed using both the segmentation loss L_{seg} on X_L and the contrastive loss L_{cont} on the whole dataset X :

$$\hat{\theta}, \hat{\xi}, \hat{\phi} = \arg \min_{\theta, \xi, \phi} \frac{1}{|X_L|} \sum_{(x_l, y_l) \in X_L} L_{seg}(y_l, \hat{y}_l) + \lambda_{cont} \frac{1}{|X|} \sum_{(x, y), (x', y') \in X} L_{cont}((x, y), (x', y'))$$

The pseudo-labels of the unlabeled images $x_u \in X_U$ are obtained as $\hat{y}_u = g_{\hat{\xi}^{(t)}}(c_{\hat{\theta}^{(t)}}(x_u))$ using the model parameters $\hat{\xi}^{(t)}$ and $\hat{\theta}^{(t)}$ at iteration t .

The proposed local pixel-wise contrastive loss L_{cont} aims to learn discriminative pixel representations based on their class ground-truth labels and pseudo-labels. To achieve this, we optimize to match the pixel representations from the same class label within the image (Intra-image matching) and across different images (Inter-image matching) to be similar while they being dissimilar to the pixel representations that belong to other classes. The total local contrastive loss between two random samples (x, y) and (x', y') from X is defined by :

$$L_{cont}((x, y), (x', y')) = \frac{1}{C} \sum_{c=1}^C \frac{1}{|S_c(x)|} \sum_{i \in S_c(x)} L_{i,c}([z(x)]_i, \bar{z}_c(x'))$$

Where $S_c(x)$ for $c \in \{1, 2, \dots, C\}$ is the set of pixel coordinates that belong to a foreground class c for image x . C is the total number of classes. $[z(x)]_i$ is the feature vector of x at pixel coordinate i . $\bar{z}_c(x') = \frac{1}{|S_c(x')|} \sum_{i \in S_c(x')} [z(x')]_i$ is the mean pixel representation of x' for class c . Finally, we can define the local contrastive loss by :

$$L_{i,c}([z(x)]_i, \bar{z}_c(x')) = -\log \frac{e^{\text{sim}([z(x)]_i, \bar{z}_c(x'))/\tau}}{e^{\text{sim}([z(x)]_i, \bar{z}_c(x'))/\tau} + \sum_{k \neq c} e^{\text{sim}([z(x)]_i, \bar{z}_k(x'))/\tau}}$$

Where sim is the cosine similarity and τ is a temperature scaling factor.

We see that, by minimizing $L_{i,c}([z(x)]_i, \bar{z}_c(x'))$, the algorithm encourages the representations $[z(x)]_i$ and $\bar{z}_c(x')$ to be similar (by maximizing the numerator) while at the same time pushing away $[z(x)]_i$ from $\bar{z}_k(x')$ for all classes k different from c , by minimizing the sum $\sum_{k \neq c} e^{\text{sim}([z(x)]_i, \bar{z}_k(x'))/\tau}$.

The authors propose two distinct configurations for the local contrastive loss to determine how feature consistency is enforced. In the Intra-image scheme, the class mean representations are computed exclusively using pixels from the same image slice as the anchor. This approach restricts the learning process to local consistency, ensuring that the model distinguishes tissues within a specific patient's scan. Conversely, the Inter-image scheme expands this scope by calculating mean representations using pixels from both the current image and other images in the training batch. By matching pixel embeddings across different subjects, the Inter-image strategy effectively forces the network to learn robust, global anatomical features that are invariant to patient-specific variations, such as differing contrast levels or scanner noise.

To assess the impact of pseudo-label quality, the authors implemented a consistency check where unlabeled volumes were subjected to geometric transformations (e.g., rotation, scaling). A pseudo-label was considered "confident" only if the Dice overlap between the masks of the same volume under two different transformations exceeded a certain threshold (e.g., 0.9). Surprisingly, strictly filtering for high-confidence labels did not yield significant performance gains compared to using all generated pseudo-labels. However, the study noted that without this filtering (threshold 0), the Inter-image matching scheme performed worse than the Intra-image scheme in low-data regimes $|X_L| \leq 2$. This suggests that attempting to match features across different subjects using noisy, low-quality pseudo-labels introduces errors that outweigh the benefits of batch diversity.

2.2 Evaluation of the segmentation performance

The authors use the Dice similarity coefficient [2] to evaluate the segmentation performance. It is defined as follows :

$$\text{DSC} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}$$

- TP (True Positive) : Pixels correctly predicted as the organ/structure.
- FP (False Positive) : Background pixels incorrectly predicted as the organ.
- FN (False Negative) : Organ pixels incorrectly predicted as background.

This DSC is averaged over all the structures in the images. This coefficient ranges from 0 to 1 and for a given segmentation result, a DSC of 1 means that there is a perfect overlap between the foreground segmented by the algorithm and the true foreground. A DSC of 0 means that there is no overlap.

3 Experiments

3.1 Datasets

The authors evaluate their model on three public datasets. In this report, we reimplemented both their methods and their model with PyTorch, and we successfully reproduced their results on the three public datasets. In addition, we introduced 2 extra datasets to further assess the robustness and generalization of their approach.

- **ACDC dataset** : 100 short-axis cardiac cine-MRI volumes acquired with 1.5T and 3T scanners, annotated for three anatomical structures : right ventricle, myocardium, and left ventricle.
- **Prostate dataset** : 48 T2-weighted MRI volumes annotated for two structures : peripheral zone and central gland.
- **MMWHS dataset** : 20 MRI volumes of the whole heart, annotated for seven cardiac structures.
- **MSD Heart (our)** : 30 heart volumes, with 2 structures : heart and background
- **CHAOS (our)** : 20 abdominal MRI volumes, annotated for four organs : liver, spleen, left kidney, and right kidney.

These datasets allow evaluation of the method in various anatomical and class-complexity scenarios.

3.2 Pre-processing

All images are first corrected using N4 bias-field correction. Each 3D volume is then normalized using min-max normalization based on the 1st and 99th percentiles. To ensure spatial consistency across subjects, all images are resampled to a fixed in-plane resolution and cropped or zero-padded to fixed dimensions. The final resolutions used are : ACDC : 192×192 , Prostate : 192×192 , MMWHS : 160×160 , Ours : MSD Heart, 192×192

3.3 Training

We use the encoder-decoder backbone described in the Methods section, along with two task-specific heads : a segmentation head and a contrastive head that produces pixel-level features.

Training also follows the two-stage strategy introduced in the Methods. We use the Adam optimizer, using a learning rate of 10^{-3} and a batch size of 20. Standard 2D medical image augmentations are applied throughout training.

For the contrastive objective, we randomly sample a small set of pixels per class and apply simple intensity perturbations, using a temperature parameter $\tau = 0.1$.

3.4 Results

3.4.1 Baselines and Compared Methods

We compare the proposed method against :

- a fully supervised baseline using only X_L ,

- self-training,
- Noisy Student,
- MixUp and data augmentation methods,
- recent semi-supervised and contrastive-learning-based approaches.

All methods use the same architecture and training schedule for fair comparison.

3.4.2 Results

We reconstructed Table 1, which reports the baseline results on the different datasets as well as the performance of the method proposed in the paper. We also extended the table by including our additional dataset.

TABLE 1 – Comparison of methods across different datasets with varying $|X_L|$ sizes

Method	ACDC			Prostate			MMWHS			MSD Heart		
	$ X_L =1$	$ X_L =2$	$ X_L =8$	$ X_L =1$	$ X_L =2$	$ X_L =8$	$ X_L =1$	$ X_L =2$	$ X_L =8$	$ X_L =1$	$ X_L =2$	$ X_L =8$
Baseline	0.614	0.702	0.844	0.489	0.550	0.636	0.451	0.637	0.787	0.533	0.629	0.664
Self-training	0.690	0.749	0.860	0.551	0.598	0.680	0.563	0.691	0.801	–	–	–
Mixup	0.695	0.785	0.863	0.543	0.593	0.661	0.561	0.690	0.796	–	–	–
Data Augment	0.731	0.786	0.865	0.585	0.597	0.667	0.529	0.661	0.785	–	–	–
Adversarial Tr.	0.536	0.654	0.847	0.487	0.544	0.586	0.482	0.655	0.779	–	–	–
Noisy Student	0.632	0.737	0.836	0.556	0.601	0.668	0.593	0.685	0.780	–	–	–
Pseudo-labels joint Tr. (intra)	0.761	0.845	0.881	0.571	0.613	0.693	0.599	0.721	0.803	0.412	0.645	0.613
Pseudo-labels joint Tr. (inter)	0.759	0.831	0.883	0.578	0.618	0.696	0.572	0.719	0.811	0.490	0.543	0.657
Training with large $ X_L $	0.912	–	–	0.697	–	–	0.787	–	–	–	–	–

Table 1 summarizes the results for different low-label regimes ($|X_L| = 1, 2, 8$). Across all datasets and supervision levels, the proposed contrastive framework achieves the highest Dice similarity coefficients. The gain is particularly pronounced in the extreme low-label setting ($|X_L| = 1$), where traditional semi-supervised methods such as self-training and Noisy Student often propagate noisy pseudo-labels, leading to unstable performance.

In contrast, semi supervised method avoids injecting pseudo-label errors into the segmentation loss and relies exclusively on them within the contrastive objective. This strategy results in more stable learning and consistently better generalization. Furthermore, inter-image contrastive matching provides stronger improvements than intra-image matching, as it captures anatomical variability across subjects; however, intra-image matching remains more robust when pseudo-labels are initially very noisy.

When evaluating our additional dataset (MSD Heart) which contains only two classes (background and heart), we observed that achieving high performance is significantly more challenging. The Dice scores remain lower compared to multi-class settings on almost all regimes

This suggests that the contrastive loss may be less effective when the segmentation task involves only two classes. The reduced variability between classes limits the benefits of inter-class contrastive separation. Consequently, naive methods such as random initialization (baseline) outperform our proposed joint method.

This difference in performance can be explained by the structure of the dataset and how we calculate the score. We report the standard "State-of-the-Art" Dice, which looks only at the heart and excludes the background.

However, if we look at the "Mean Dice" (which includes the large background), our results are actually very good, reaching scores above 0.85. The drop in performance only appears when we focus strictly on the small heart structure.

Interestingly, on this specific dataset, the standard supervised baseline performs better than our joint training method. We believe this is because the joint method relies on a contrastive loss, which usually needs multiple classes to be effective. With only one object (the heart) versus a large background, the contrastive loss seems to struggle, whereas the simple baseline handles this strong imbalance more effectively.

3.4.3 Ablation Studies

They conduct extensive ablations to assess the influence of key components :

- weight λ_{cont} of the contrastive loss,
- number of pseudo-labeling iterations,
- number of sampled pixels per class,
- filtering pseudo-labels based on high-confidence criteria.

The ablation study shows that each component of the framework contributes to the overall performance gains. The weight of the contrastive loss, the number of pseudo-labeling iterations, and the number of sampled pixels per class all influence the results, yet the method remains generally robust to these variations. Finally, filtering pseudo-labels based on high-confidence criteria does not yield noticeable improvements, indicating that using all pseudo-labels directly within the contrastive loss is sufficient to stabilize and strengthen the learning process.

3.5 Visualisation analysis

3.5.1 Slices examples

We introduce the CHAOS dataset in this experiment to provide a setting with multiple abdominal organs, allowing us to explore and visualize the model’s behavior in a richer multi-class context. It contains healthy abdominal MRI scans with annotations for the liver, spleen, and both kidneys.

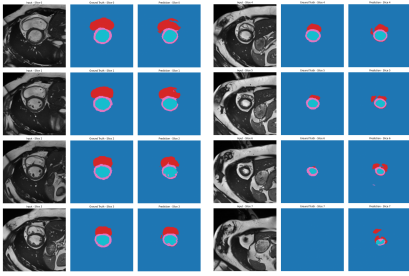


FIGURE 3 – ACDC slices

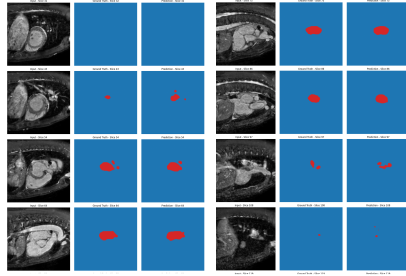


FIGURE 4 – MSD Heart slices

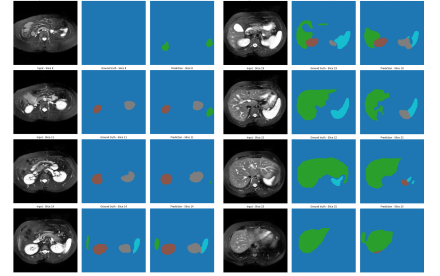


FIGURE 5 – CHAOS slices

For comparison, we also visualize slices from the ACDC, MSD Heart, and CHAOS datasets. In ACDC, some cardiac substructures are successfully segmented while others remain partial, reflecting an average Dice score of around 0.79. In MSD Heart, which contains only two classes (heart vs. background), the segmentation is more consistent and complete, with a Dice score of approximately 0.88. Finally, CHAOS includes multiple abdominal organs, with generally correct segmentations but greater complexity, highlighting the challenges of multi-class segmentation and the richness of anatomical structures.

3.5.2 Visualisation t-SNE

To better understand how the model structures its latent space, we computed t-SNE projections of the decoder features on three different datasets : ACDC, MSD Heart, and CHAOS.

These datasets differ in the number of anatomical structures they contain, which allows us to analyze how the latent representations behave as the number of classes increases.

On the ACDC dataset, which contains several cardiac substructures, the t-SNE shows a clear pattern of compartmentalization : each class tends to form its own region in the latent space, with relatively well- defined boundaries between them. This indicates that the model is able to separate multiple foreground classes, and the latent geometry reflects the multi-organ nature of the task.

On the MSD Heart dataset, the situation is different due to the strong class imbalance between the single organ (heart) and the background. As shown in the t-SNE visualization, the background forms

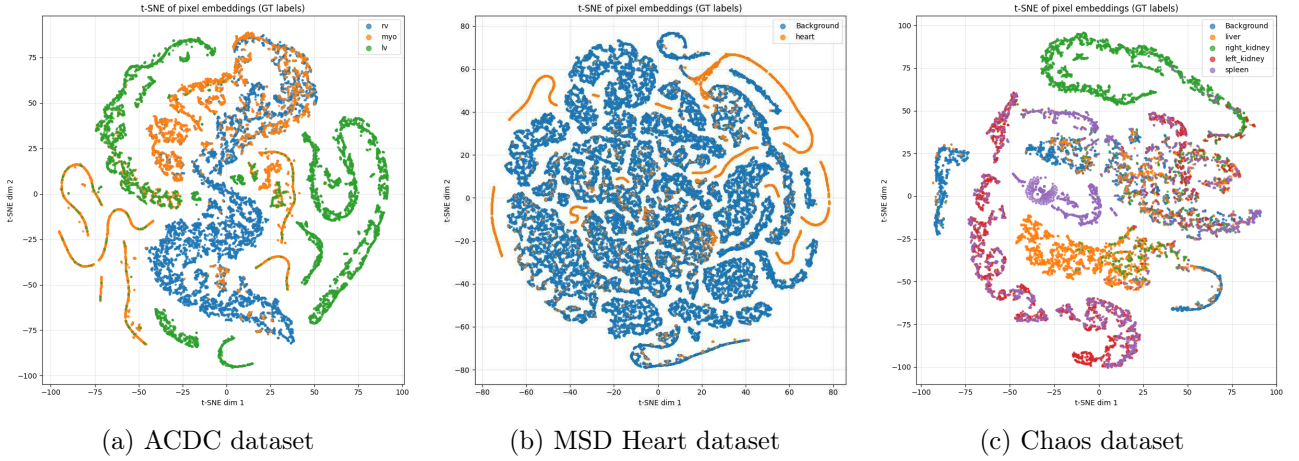


FIGURE 6 – t-SNE of latent space for different datasets

a large, complex cluster due to its high visual variability. In contrast, the heart voxels cluster into elongated, thin structures. While some of these structures are distinct, we observe that several heart points blend directly into the dense background clusters.

This partial overlap and extreme proximity indicate a narrow margin of separation in the feature space. With only one foreground class, the contrastive loss establishes a decision boundary that remains fragile. Consequently, while the model learns the global structure, it struggles to achieve the robust separation seen in multi-class datasets, explaining why the foreground-only Dice score drops compared to the supervised baseline.

Finally, on the CHAOS dataset, which includes several abdominal organs, we observe once again a much more structured and well-separated t-SNE map. The latent space contains multiple clearly defined clusters, each corresponding to a different organ. In this multi-class setting, the joint objective becomes far more effective, as there are meaningful relationships between multiple foreground classes for the model to organize. The separation improves, and the latent representations appear significantly richer.

Overall, these experiments show that the benefit of the joint loss increases with the number of classes. The more organ classes the dataset contains, the more compartmentalized and interpretable the latent space becomes. Conversely, datasets with only one foreground class, such as MSD Heart, provide limited structure for the contrastive loss, leading to less informative feature organization.

4 Discussion - Conclusion

4.1 State of the Art

Although the paper Local Contrastive Loss with Pseudo-Label Based Self-Training (2021) introduced an effective framework for semi-supervised medical image segmentation, the field has evolved significantly over the last few years. With the emergence of Transformer-based architectures, multi-scale representation learning, and more advanced contrastive objectives, several recent methods have surpassed the performance of local-contrastive approaches.

In particular, the work Multi-Scale Cross Contrastive Learning for Semi-Supervised Medical Image Segmentation (Liu et al., 2023) [3] proposes a more powerful framework combining CNN and Transformer backbones with multi-scale contrastive learning and cross-pseudo supervision. Instead of focusing only on local relationships, as in the 2021 method, Liu et al. enforce consistency and semantic alignment across multiple feature resolutions and between heterogeneous models. This allows their system to capture both fine-grained local details and long-range anatomical dependencies, leading to more discriminative and robust representations even with very limited labeled data.

As a result, their model consistently outperforms earlier contrastive-pseudo-label approaches. For example, on ACDC, with only $X_L = 7$ labeled volumes, their method achieves higher segmentation dice than the local-contrastive framework using $X_L = 8$ labeled volumes they achieved a dice score of 89.4 while we had 88.3.. This demonstrates that recent multi-scale and Transformer-augmented methods have become the new state of the art (SOTA) in semi-supervised medical image segmentation, pushing beyond the capabilities of earlier local-contrastive approaches.

Labelled	Methods	Mean		Myo		LV		RV	
		DSC↑	HD↓	DSC↑	HD↓	DSC↑	HD↓	DSC↑	HD↓
70 cases (100%)	UNet-FS	91.7	4.0	89.0	5.0	94.6	5.9	91.4	1.2
	BATFormer [16]	92.8	8.0	90.26	6.8	96.30	5.9	91.97	11.3
7 cases (10%)	UNet-LS	75.9	10.8	78.2	8.6	85.5	13.0	63.9	10.7
	CCT [19]	84.0	6.6	82.3	5.4	88.6	9.4	81.0	5.1
	CPS [8]	85.0	6.6	82.9	6.6	88.0	10.8	84.2	2.3
	CTS [17]	86.4	8.6	84.4	6.9	90.1	11.2	84.8	7.8
	MCSC (Ours)	89.4	2.3	87.6	1.1	93.6	3.5	87.1	2.1

FIGURE 7 – Multi-Scale Cross Contrastive Learning for Semi-Supervised Medical Image Segmentation on ACDC

4.2 Conclusion

In this project, we analysed and reproduced the framework of Local Contrastive Loss with Pseudo-Label based Self-Training for semi-supervised medical image segmentation. The key idea is to move from global image-level contrast to a pixel-wise (local) contrastive loss. This local objective enforces anatomical consistency and helps the network separate fine structures even when the number of labeled scans is very small ($X_L \leq 8$).

On the ACDC dataset, our experiments confirm the findings of the original paper : the local contrastive loss clearly outperforms standard self-training and consistency-based baselines. It reduces the propagation of noisy pseudo-labels and leads to more stable training in the low-data regime.

We then extended the study to two additional datasets : CHAOS (multi-organ, five classes) and MSD Heart (one organ class and background). The MSD Heart experiments revealed an important limitation of the framework. Using t-SNE visualisations, confusion matrices and per-patient Dice scores, we observed that the contrastive loss is much less effective in binary or highly unbalanced settings (one small foreground class vs. a very large background). In this case, there are not enough push / pull constraints between different organs in the latent space, and the features remain less discriminative. On MSD Heart, the supervised baseline often performs as well as, or better than, the joint training. In contrast, on ACDC and CHAOS, where several organ classes are present and the background is less dominant, the joint training with local contrastive loss consistently improves over the baseline.

Finally, while this method was a strong contribution in 2021, our short literature review shows that the field has moved towards Transformer-based and multi-scale architectures. Recent approaches, such as Liu et al. (2023) [3], obtain higher performance by capturing long-range dependencies that a purely CNN-based contrastive model cannot fully exploit. Nevertheless, the core idea of local pixel-level contrast remains important : enforcing local semantic consistency is a powerful and still relevant principle for robust semi-supervised medical image segmentation.

Références

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